

Salvador de Sousa Carvalho

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ABOUT

I am a 3rd-year Electrical and Computer Engineering student at **Instituto Superior Técnico** focused on developing my skills both on hardware and software. My entire life has been marked for a long standing curiosity for knowing how things work, it all began with simple sets of Lego Technic and evolved to more technical topics like **Robotics, IoT, and Home Automation**.

With a **Cambridge C1 Advanced** certification and currently learning German, I am eager to keep learning and pushing myself into new challenges and to apply what I learn to new projects.

EXPERIENCE

Formula Student Lisboa, Lisbon, Portugal Jan 2025 – Aug 2025
Trainee, Electronics and Software Department

- Designed and developed electronic schematics and PCB layouts for the vehicle's braking system sensors using **Altium Designer**.
- Programmed microcontrollers in **C** and **C++** to process sensor data and implement real-time control logic for the braking system.
- Performed high-precision **soldering** and hardware assembly, ensuring the reliability of electronic components in high-performance environments.
- Collaborated in team communication and technical presentations, effectively reporting progress and system design specifications.

EDUCATION

Instituto Superior Técnico, Lisbon, Portugal Sep 2023 – Present
Bachelor of Science in Electrical and Computer Engineering

Colégio de São Miguel, Fátima, Portugal Sep 2020 – Jun 2023
Secondary Education, Ciências e Tecnologias **GPA: 17.2/20**
National Mathematics Exam: 18.1/20

PROJECTS

Forest Fire Detection System (PIC) In Development

- Developed an integrated monitoring system for early fire detection using a **YOLO-based** AI model for real-time smoke identification via camera modules.
- Integrated environmental sensors with **Meshtastic (LoRa)** to enable long-range, off-grid communication in remote forest areas.
- Built a custom **unified dashboard** to aggregate sensor telemetry and AI visual alerts into a single, real-time control interface.

Distributed Resource Sharing System (RC) ([GitHub](#)) In Development

- Engineered a distributed application in **C** using a client-server and peer-to-peer architecture.
- Implemented reliable data transfer and discovery protocols using **TCP and UDP socket programming**.
- Developed a concurrent server capable of handling multiple simultaneous requests with robust error handling.

Image Watermark & PDF Processor([GitHub](#)) 2025

- Developed a **React** web application for processing JPEG images with custom watermarks.

- Integrated **Canvas API** for dynamic image manipulation and **jsPDF** for secure PDF generation.
- Implemented 100% client-side processing to ensure data privacy and responsive UI with **Tailwind CSS**.

Algorithms and Data Structures Projects ([GitHub](#))

2025

- Developed a series of applications in **C** focused on algorithmic efficiency and memory management. Implemented data structures including **Min-Heaps**, **Linked Lists**, and **Hash Tables**, and applied graph algorithms such as **Dijkstra** for shortest-path problems. Focused on optimizing time and space complexity (O -notation) to handle large datasets.

Image Watermark & PDF Processor([GitHub](#))

2025

- Developed a **React** web application for processing JPEG images with custom watermarks.
- Integrated **Canvas API** for dynamic image manipulation and **jsPDF** for secure PDF generation.
- Implemented 100% client-side processing to ensure data privacy and responsive UI with **Tailwind CSS**.

CS2 & Faceit Discord Bot ([GitHub](#))

2025

- Developed a custom Discord bot using **discord.py** and **aihttp** to integrate Faceit API data and live CS2 server status via the A2S protocol. Implemented real-time Elo tracking, match statistics retrieval, and an interactive server browser menu, alongside a voice-system module utilizing **FFmpeg** and **PyNaCl** for custom audio playback.

5-DOF Robotic Arm ([GitHub](#), [Video](#), [Report](#))

2023

- Developed a 5-DOF robotic arm control system using Arduino and C++, featuring a Serial interface for real-time joint manipulation and user interaction. Implemented input validation and safety constraints for servo positioning to ensure smooth kinematic movement and hardware longevity. Documented the complete electronic circuit design and firmware architecture in a detailed technical report.

HONORS & AWARDS

Merit Board , Colégio do Sagrado Coração de Maria	2017/18, 2018/19, 2019/20
Honor Board , Colégio do Sagrado Coração de Maria	2017/18, 2018/19, 2019/20
Merit Board , EB1 Reguengo do Fetal	2013/14

CERTIFICATES

Cambridge Advanced English , Cambridge English	Aug 2023
Cambridge First Certificate in English , Cambridge English	Jul 2022

SKILLS AND INTERESTS

Hard skills:

- Programming: C/C++, Python, MATLAB, Simulink, Fusion360
- Tools & Frameworks: Git, Docker, Linux, LTSPICE

Soft skills: Analytical & Creative Thinking, Team Collaboration, Adaptability & Communication

Language skills: Portuguese (native), English (C1), German (A1)